

Counts or Conditional Percents?

Unit 2 · Topic 2.1 · THE PROBLEM

Suppose a counselor records whether each of 200 students plays a school sport and whether the student usually gets at least 8 hours of sleep on a school night.

Kai: *Seventy-five nonathletes but only 35 athletes get at least 8 hours, so playing a sport goes with less sleep.*

Rosa: *Those counts compare groups of different sizes; 35 out of 50 may tell a different story from 75 out of 150.*

Usual sleep on a school night

Sport?	≥ 8 h	< 8 h	Total
Plays	35	15	50
Does not	75	75	150
Total	110	90	200

FIRST TAKE — one sentence before you work the parts

Which student's reasoning seems more useful for deciding whether sport participation and sleep are related? Name one table detail.

DOK 1 (a) Name the two categorical variables.

DOK 2 (b) Find the percent who get at least 8 hours of sleep in each sport group. Show the two fractions.

DOK 3 (c) Whose reasoning is better supported? Use both percentages and explain why comparing 75 with 35 can mislead. State whether this table shows that playing a sport causes more sleep. Write 3–4 sentences.

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.